INTERNATIONAL news

Israel announces $3.2B investment in Intel's chip plant near Gaza

The Israeli government has announced a significant investment of $3.2 billion for Intel's new $25 billion chip plant in southern Israel, 42 kilometres from Gaza. The chip plant, named Fab 38, is scheduled to open in 2028 and operate until 2035.

Renesas acquires Transphorm for GaN tech expansion

Renesas Electronics Corporation has acquired GaN based devices producer Transphorm for $339 million. The acquisition allows Renesas to integrate GaN technology, crucial for the next-generation power semiconductors, thereby enhancing its presence in rapidly growing markets like electric vehicles, data centres, AI, renewable energy, and industrial power conversion. Amid the shift towards wide bandgap (WBG) materials like silicon carbide (SiC) and GaN, Renesas is also setting up an in-house SiC production line and has secured a decade-long SiC wafer supply.

US EV policy likely to continue reliance on FTA nations for battery metals

The US Electric Vehicle (EV) policy, aiming for 67% EVs by 2032, emphasises domestically sourced battery metals, but limited production capacity may sustain reliance on international suppliers. Free Trade Agreement (FTA) nations play a crucial role in shaping the EV supply chain, with strategic mineral offtake agreements gaining traction.

Meta to shift to RISC-V architecture to bypass reliance on Nvidia GPUs

Meta is making a strategic shift towards the open source RISC-V architecture for AI acceleration, replacing the traditional x86 architecture, aiming for increased accessibility, cost-effectiveness, and efficiency in AI applications. Meta aims to leverage RISC-V for AI training and inference, bypassing reliance on Nvidia GPUs. Meta has already implemented RISC-V in its video transcoding hardware, Meta scalable video processor (MSVP), replacing 85% of previously used CPUs.

Chip giants establish Quintauris to promote RISC-V

Headquartered in Munich, Germany, semiconductor players formally established Quintauris GmbH to catalyse RISC-V adoption. The company is a result of collaboration between Robert Bosch GmbH, Infineon Technologies AG, Nordic Semiconductor ASA, NXP Semiconductors, and Qualcomm Technologies, Inc. Quintauris' primary goal is to serve as a central resource in promoting RISC-V-based products. The company will provide reference architectures and develop solutions applicable across various industries, starting with automotive and eventually expanding into mobile and the internet of things (IoT) sectors.

Thailand investment board plans enhanced support for domestic electronic circuit industry

Thailand's Board of Investment (BOI) aims to boost the local electronic circuit industry by offering support and incentives to position the country as Southeast Asia's electronic industry hub. The focus will be on the advanced production of critical components like printed circuit boards and printed circuit board assemblies.

Microchip Technology secures funding to boost US domestic chip production

Microchip Technology is getting $162 million from the US government. This funding aims to boost domestic production of computer chips and microcontroller units. The fund comprises $90 million designated for enhancing a chip fabrication plant in Colorado and an additional $72 million for expanding a second facility in Oregon. It will help Microchip triple the production of its mature-node semiconductor chips and microcontroller units at two factories in the US.

South Korea unveils plan to build 'semiconductor mega cluster' by 2047

South Korea plans to build the world's largest semiconductor mega cluster in Yongin, southern Seoul by 2047. This cluster will cover numerous industrial zones throughout southern Gyeonggi Province, boasting a total area of 21 million square metres. Thirteen new plants will be built with a monthly production capacity of 7.7 million wafers by 2030. It will create at least 3 million jobs over the next 20 years, with the support of Samsung Electronics Co. and SK hynix Inc. with an investment of 622 trillion. South Korea is already home to 21 fabrication plants.

Samsung Electronics to invest in Japan chip packaging research plant

Samsung Electronics plans to invest around $280 million (40 billion yen) over five years in a chip packaging research facility in Japan, coinciding with improved relations between South Korea and Japan. The facility will focus on research into advanced chip packaging and will be established in Kanagawa prefecture, which already houses a research and development centre. Japan's industry ministry will provide up to 20 billion yen in subsidies to Samsung, supporting the revitalisation of domestic chip manufacturing.

Netherland denies China ASML's advance lithography systems

The Netherlands has partially revoked ASML's export license for advanced lithography systems bound for China, impacting ASML's sales and drawing criticism from Beijing. The move is influenced by US government pressure to limit China's ability to manufacture advanced chips. The affected machines required a Dutch licence for export to China since September 2023, following US regulations.

IEC and ISO establish committee for quantum technology

The International Electrotechnical Commission (IEC) and the International Organization for Standardization (ISO) have jointly established a new technical committee dedicated to quantum technologies, named IEC/ISO JTC 3: Quantum technologies. The joint technical committee will focus on developing standards in the field of quantum technologies, specifically in areas such as quantum computing, quantum simulation, quantum sources, quantum metrology, quantum detectors, and quantum communications.

Microsoft and Vodafone to offer generative AI, digital, enterprise, and cloud services in Europe and Africa

Vodafone has entered a 10-year partnership with Microsoft to offer generative AI, digital, enterprise, and cloud services to over 300 million businesses and consumers across Europe and Africa. The British company plans to invest $1.5 billion in AI, using Microsoft's Azure OpenAI and Copilot technologies. This investment will also involve transitioning from physical data centres to Azure cloud services. The objective is to build digital literacy and introduce AI for intelligent financial decision-making.